

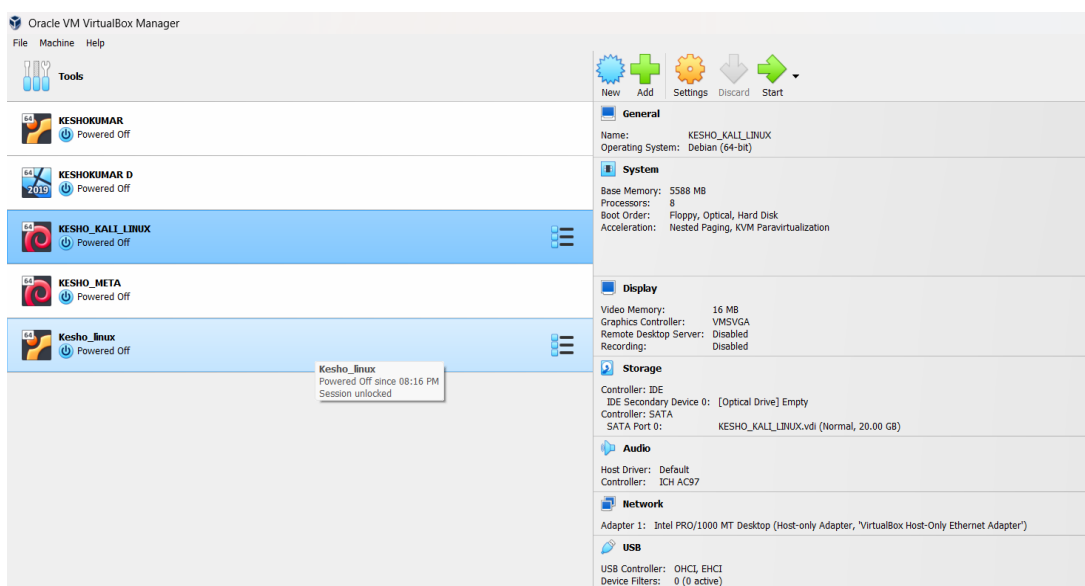
2a) Creating a VM and adding Storage, Memory, and Network to a VM

Aim:

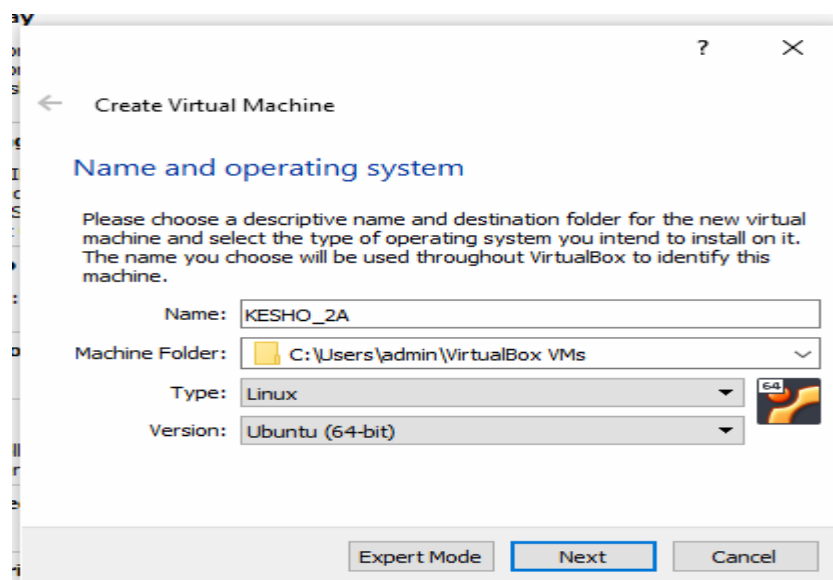
To create a VM and adding storage ,memory and network to a VM in the oracle virtual box

Steps:

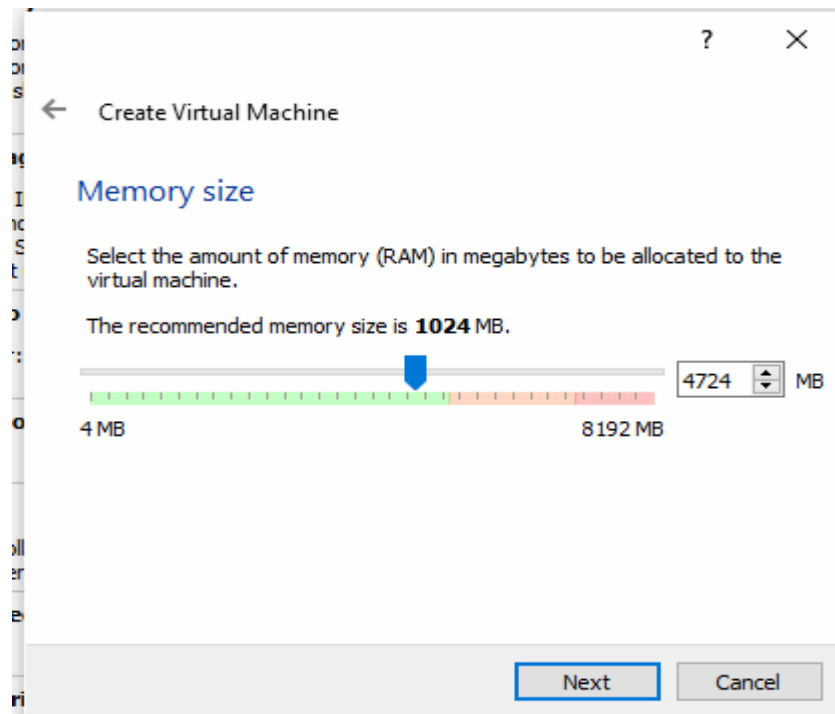
- 1) Open Oracle VM VirtualBox Manager and Click the "New" button



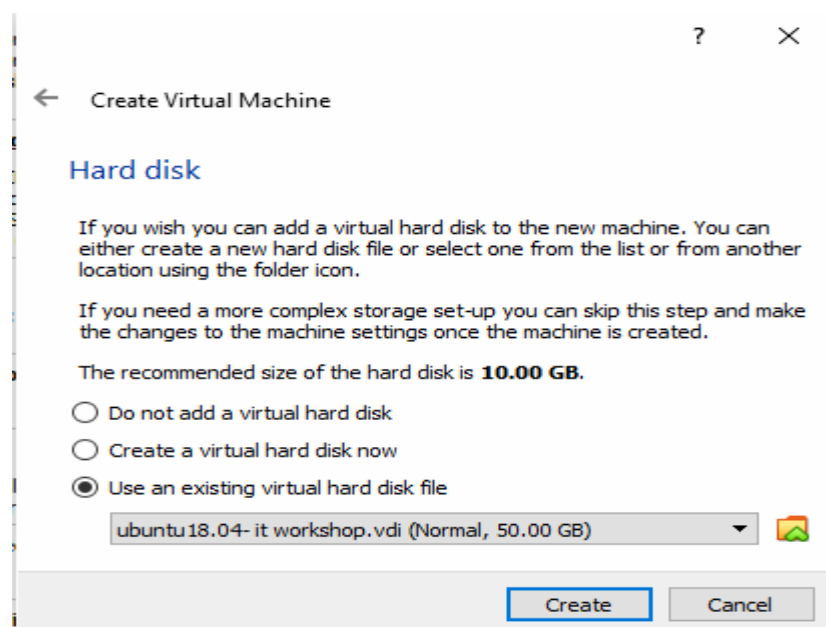
- 2) Fill the name: (KESHO_2A) Type: Linux Version: Ubuntu (64-bit) or your chosen OS and Click Next



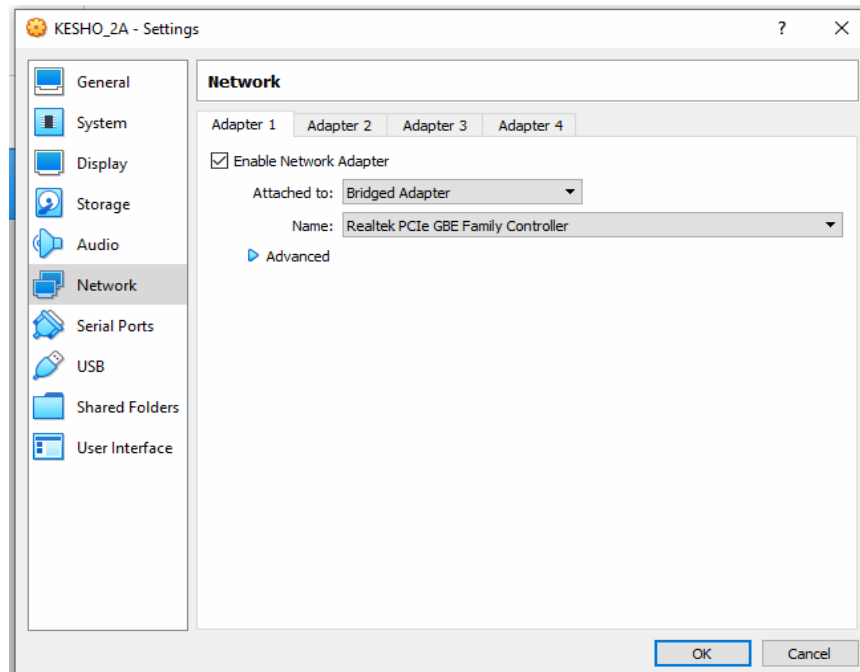
3) Allocate Memory (RAM) .Set the memory size 4gb has been set below Don't exceed 50% of your physical system RAM Click Next



4)Click use an existing virtual hard disk. Click Next Dynamically allocated Click Next Set disk size (e.g., 20 GB)Click Create



5) Configure Network Adapter Go to Settings Network Ensure Adapter 1 is: Enabled Attached to: Bridged Adapter (for internet + LAN access), or Host-Only Adapter (for communication with host only) Click OK Then new vm has been created



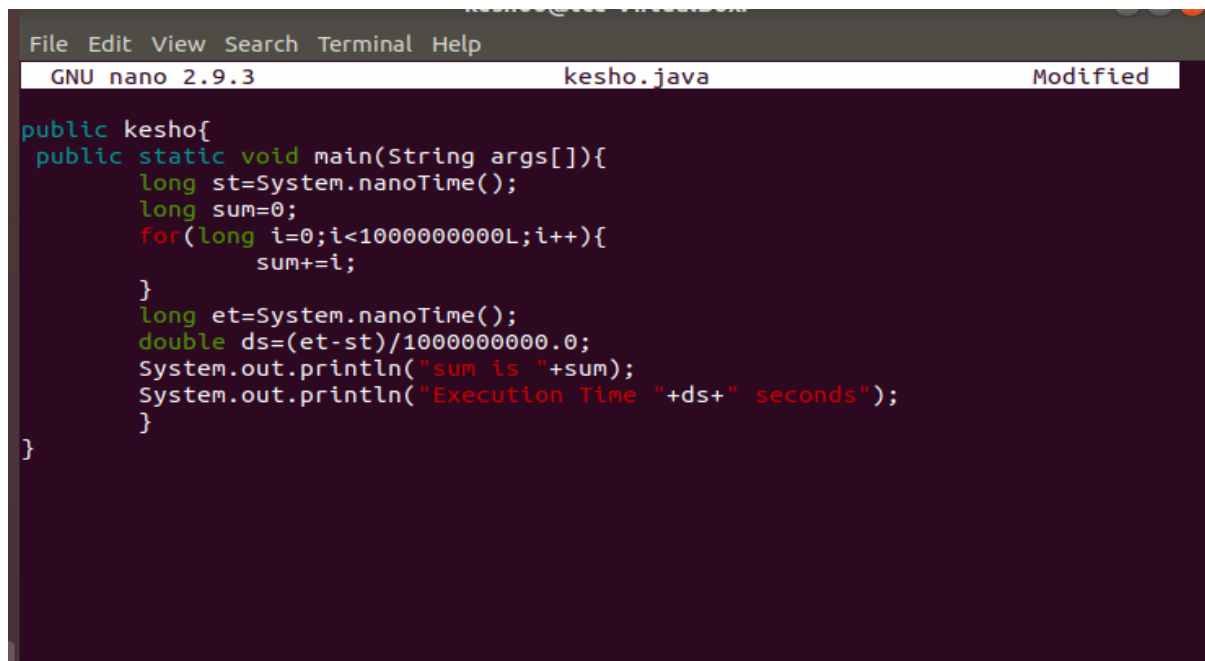
Execution of java program to test the execution time between vm (Ubuntu) and host(Windows)

1)To find the sum of digits between 1 to max of long datatype

Steps to Execute in Ubuntu (Virtual Machine)

1. Open the Terminal.
2. Create a Java source file using the command:
`gedit filename. java`
3. Type the Java program in the Gedit editor.
4. Save and close the editor.
5. Compile the program using:
`javac filename.java`
6. Execute the program using:
`java filename`

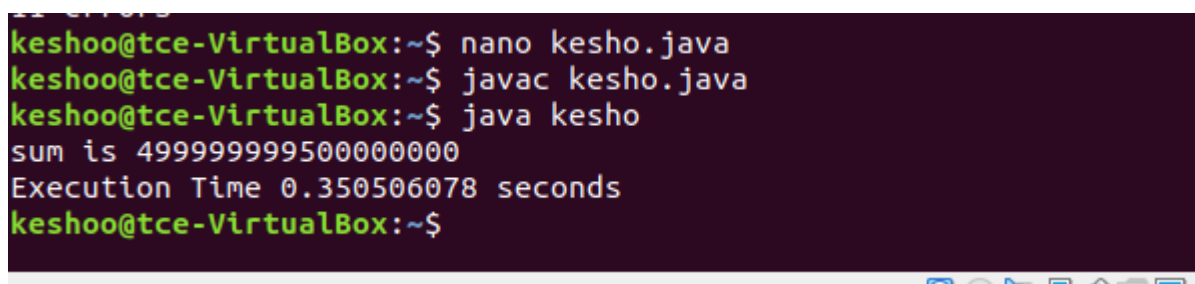
Code:



```
File Edit View Search Terminal Help
GNU nano 2.9.3 kesho.java Modified

public kesho{
  public static void main(String args[]){
    long st=System.nanoTime();
    long sum=0;
    for(long i=0;i<10000000000L;i++){
      sum+=i;
    }
    long et=System.nanoTime();
    double ds=(et-st)/1000000000.0;
    System.out.println("sum is "+sum);
    System.out.println("Execution Time "+ds+" seconds");
  }
}
```

Execution time in Linux



```
keshoo@tce-VirtualBox:~$ nano kesho.java
keshoo@tce-VirtualBox:~$ javac kesho.java
keshoo@tce-VirtualBox:~$ java kesho
sum is 499999999500000000
Execution Time 0.350506078 seconds
keshoo@tce-VirtualBox:~$
```

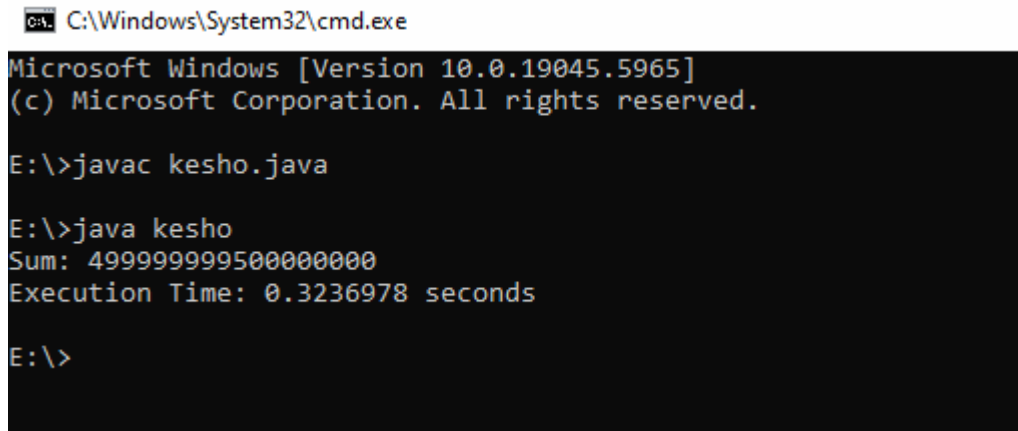
Execution time in Windows

Steps:

1. Open notepad in windows.
2. Type the program and save it as filename.java
3. Open Command Prompt in windows
4. Navigate to the folder containing the Java program:
cd <path_to_program> example:D:/
5. Compile the program: javac filename.java
6. Run the program: java filename

DATE :

Reg.no_student name



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

E:\>javac kesho.java

E:\>java kesho
Sum: 499999999500000000
Execution Time: 0.3236978 seconds

E:\>
```

From the above the execution time in **linux is 0.35 s** where the time taken by the **windows is 0.32** this clearly tells that the windows host is faster than the linux vm

RESULT:

Thus the Creation of VM and adding Storage, Memory, and Network to a VM has been created successfully and output has been verified

2b) VM-VM ping, VM-Native Ping

Aim:

To check the victim or target is alive between VM -VM and VM-native (host) using the ping command .

Steps: Before ping commands

- Both VMs are Powered ON
- Both are connected to the same network (e.g., Host-Only or Bridged)
- Each VM has a valid IP address (like 192.168.56.x)
- Firewall is disabled or ICMP is allowed

Find IP Addresses

- 1) Using ip a or ifconfig in VM
- 2) Using ipconfig in windows

To enable VM-to-VM and VM-to-host ping communication in VirtualBox, first set both VMs' network adapters to "Host-Only Adapter" (e.g., vboxnet0) via VirtualBox Settings Network. Then, start both VMs and run ip a inside each to find their IP addresses (usually 192.168.56.x). On each VM, disable the firewall using `sudo ufw disable` and optionally flush iptables with `sudo iptables -F`. On the Windows host, enable ICMP ping by opening "Windows Defender Firewall with Advanced Security", going to Inbound Rules, and enabling "File and Printer Sharing (Echo Request - ICMPv4-In)". Now, use `ping <other_VM_IP>` from one VM to the other, and from the host, use Command Prompt to ping each VM's IP (e.g., `ping 192.168.56.101`)

1)VM – NATIVE:

To check the target is alive between the host and vm using ping commands

IP address of windows:

DATE :

Reg.no_student name

Ip address of linux was 127.0.0.1

Using ping checking the vm is alive or not (ping 127.0.0.1)

```
C:\Users\admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::31f5:3981:81a0:e74f%19
    IPv4 Address. . . . . : 10.10.25.122
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 10.10.24.1

Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::202:e468:cccc:6575%18
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\Users\admin>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Ping for windows:

Using ping checking the host is alive or not (ping 127.0.0.1)

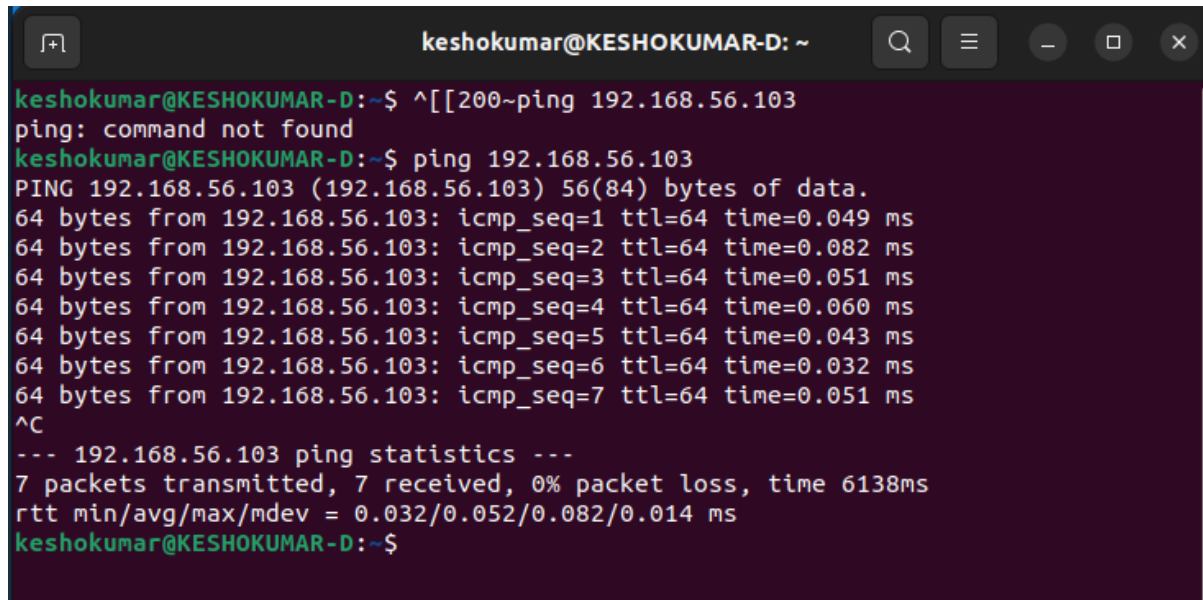
```
File Edit View Search Terminal Help
inet6 fe80::9649:f263:2e81:6faa/64 scope link noprefixroute
valid_lft forever preferred_lft forever
keshoo@tce-VirtualBox:~$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.019 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.059 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.129 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.293 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.064 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.121 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.058 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.020 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.064 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.032 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.026 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.024 ms
64 bytes from 127.0.0.1: icmp_seq=15 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=16 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=17 ttl=64 time=0.130 ms
64 bytes from 127.0.0.1: icmp_seq=18 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=19 ttl=64 time=0.040 ms
^C
--- 127.0.0.1 ping statistics ---
19 packets transmitted, 19 received, 0% packet loss, time 18424ms
rtt min/avg/max/mdev = 0.019/0.072/0.293/0.062 ms
keshoo@tce-VirtualBox:~$
```

VM- VM

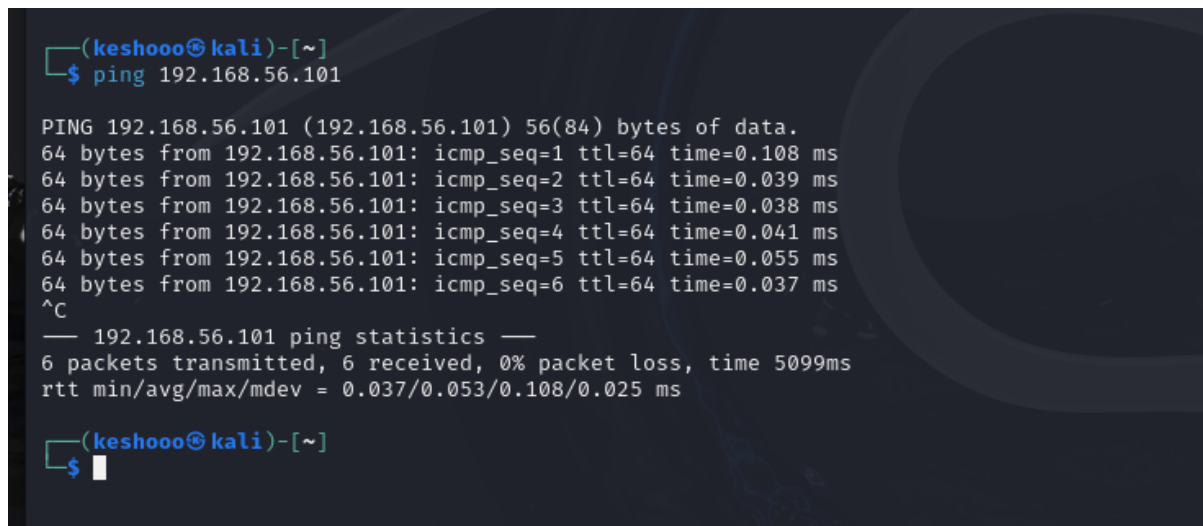
I am using linux (one was ubuntu) and another was (Kali linux)

Ping in Linux ubuntu:

Using ping 192.168.56.103 (Checking the kali linux is alive or not using ping in ubuntu)

A terminal window titled 'keshokumar@KESHOKUMAR-D: ~' with standard window controls. The terminal shows a user attempting to ping 192.168.56.103. The first attempt fails with 'ping: command not found'. The second attempt succeeds, showing 7 successful pings with varying response times. The user then presses Ctrl+C to stop the command, which displays ping statistics: 7 packets transmitted, 7 received, 0% packet loss, and a total time of 6138ms. The terminal ends with the prompt 'keshokumar@KESHOKUMAR-D:~\$'.

Using ping 192.168.56.101 (Checking the ubuntu linux is alive or not using ping in kali linux)

A terminal window titled '(keshoooo@kali)-[~]' with standard window controls. The terminal shows a user attempting to ping 192.168.56.101. The command succeeds, showing 6 successful pings with varying response times. The user then presses Ctrl+C to stop the command, which displays ping statistics: 6 packets transmitted, 6 received, 0% packet loss, and a total time of 5099ms. The terminal ends with the prompt '(keshoooo@kali)-[~]\$'.

RESULT

Thus the checking the victim or target is alive between vm -vm and vm-native using the ping command has been executed successfully and output has been verified.